

Berge–Fulkerson Covers, Cycle Double Covers, Flows and Exact Normality Defects of the First Counterexamples to the Petersen Coloring Conjecture

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CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

18 September 2026

Preprint (independent certification and consequence audit)

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CAOS Research open-problems programme · problem `petersen-coloring`

Version DOI [10.5281/zenodo.22823416](https://doi.org/10.5281/zenodo.22823416)

Concept DOI (always latest) [10.5281/zenodo.22285164](https://doi.org/10.5281/zenodo.22285164)

Not peer reviewed. Source code, data and computational records:

github.com/fsantibanezleal/CAOS_RESEARCH

Abstract

Jaeger’s Petersen coloring conjecture (1988), which asserted that every bridgeless cubic graph admits an edge map onto the Petersen graph sending every vertex star onto a vertex star, was refuted in August 2026: Putman exhibited two nonisomorphic 112-vertex counterexamples with SAT-solver certificates, Jookan gave a human-checkable proof, and Goedgebeur, Jookan, Máčajová, Mattiolo and Mazzuoccolo found a 52-vertex counterexample together with infinite families, placing the smallest counterexample between 38 and 52 vertices. The conjecture had been the strongest known sufficient condition for the Berge–Fulkerson conjecture and the 5-cycle double cover conjecture. We audit, by exact proof-carrying computation, what survives on the three retrievable counterexamples. First, independent encodings sharing no variable scheme with the public ones refute Petersen colorings and normal 5-edge-colorings of all three graphs with proofs checked by `drat-trim`; five colorable controls are accepted, and Putman’s four public proofs verify under our checker. Second, every one of the three graphs admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, a Fan–Raspaud triple, a 5-cycle double cover and a nowhere-zero 5-flow, all given as explicit witnesses re-verified from the graph alone; none admits a nowhere-zero 4-flow. Their perfect matching index is exactly 4, one below the Petersen graph’s value. The two 112-vertex graphs have oddness 4 and resistance 3, whereas the 52-vertex graph has oddness 2 and resistance 2, every value certified from both sides. All three admit normal and strong normal 6-edge-colorings, so their normal chromatic index is exactly 6. Defining the Petersen defect as the least number of vertices at which the star condition must fail, we prove that the 52-vertex graph has defect exactly 2: no single vertex absorbs the obstruction (52 checked refutations), while every one of its 1,326 vertex pairs does (explicit witnesses); the same holds for the two 112-vertex graphs at every one of their 6,216 pairs. A parity argument explains the first half in general: for any edge map into the Petersen graph the number of bad vertices is never exactly one, so every counterexample has defect at least 2 and

the three known counterexamples are extremal. Finally, we prove that no counterexample consists solely of copies of the Petersen 4-pole F from which all known counterexamples are built, and report a counterexample-guided search over compositions of F with free vertices below 52 vertices. All computations are propositional with checked DRAT certificates or explicit witnesses. The paper decides nothing about the general conjectures.

1 Introduction

1.1 The conjecture and its ladder

Let P denote the Petersen graph. For a cubic graph G , a *Petersen coloring* is a map $\sigma : E(G) \rightarrow E(P)$ such that for every vertex v of G there is a vertex w of P with $\sigma(\partial_G(v)) = \partial_P(w)$, where $\partial(\cdot)$ denotes the set of edges at a vertex; equivalently, the three edges at v map bijectively onto the three edges at w . A proper 5-edge-coloring is *normal* if every edge uv is *poor* ($|c(\partial u) \cup c(\partial v)| = 3$) or *rich* ($= 5$); the *normal chromatic index* $\chi'_N(G)$ is the least k admitting a normal k -edge-coloring.

Conjecture 1.1 (Jaeger 1988 [1]). *Every bridgeless cubic graph admits a Petersen coloring.*

Jaeger [2] proved that a cubic graph has a Petersen coloring iff it has a normal 5-edge-coloring (an explicit bijective form is Proposition 2.1 of [3]). The conjecture implies the Berge–Fulkerson conjecture (six perfect matchings covering every edge exactly twice) and the 5-cycle double cover conjecture [4, 6]; the Berge conjecture (perfect matching index at most 5) and the Fan–Raspaud conjecture (three perfect matchings with empty intersection) follow from Berge–Fulkerson. Every 3-edge-colorable cubic graph is trivially Petersen colorable, so the whole ladder concerns snarks. Exhaustive generation had verified the conjecture for every snark on at most 36 vertices [7] and for every weak snark on 36 vertices [8], so any counterexample has at least 38 vertices [5].

1.2 The disproof

On 6 August 2026 Putman [3] published a simple bridgeless cubic graph G_{112} on 112 vertices and 168 edges with no Petersen coloring and no normal 5-edge-coloring, certified by CaDiCaL proofs checked with `drat-trim`, together with a nonisomorphic D_3 -symmetric counterexample H_{112} ; both have girth 5 and edge- and vertex-connectivity 3. The construction composes copies of the 4-pole F (the Petersen graph minus the endpoints of one edge) with claw six-poles: $L = 4F + C$ and $G_{112} = 3L + C$. Jookan [4] gave a human-checkable proof through the Petersen-coloring sets of F and L and a 4-clique obstruction in the line graph of P . Goedgebeur, Jookan, Máčajová, Mattiolo and Mazzuoccolo [5] then gave a cyclically 4-edge-connected girth-5 counterexample G_{52} on 52 vertices, infinite cyclically 4-edge-connected families, and the frontier $38 \leq n_{\min} \leq 52$ for the smallest counterexample; they verified a strong normal 6-edge-coloring of one 112-vertex graph and left open whether every bridgeless cubic graph has a normal 6-edge-coloring (their Conjecture 6, attributed to Šámal) and whether cyclically 5-edge-connected counterexamples exist. A 68-vertex counterexample was announced on a social network without a retrievable edge list and is not treated here.

1.3 What this paper does

The three retrievable counterexamples are the first bridgeless cubic graphs for which the Petersen route to a Berge–Fulkerson cover or a 5-cycle double cover is closed. We therefore ask, for each of G_{112} , H_{112} and G_{52} , which of the conjectures below the top of the ladder still hold, and how far each graph is from being colorable. Every answer is exact: a positive answer is an explicit witness re-verified by a standard-library checker that reads only the graph, and a negative answer is a DRAT proof checked by `drat-trim`. Section 2 gives the independent certification (a replication of the published refutations), Section 3 the perfect matching covers, Section 4 the cycle double covers, flows, oddness and resistance, Section 5 the normal 6-edge-colorings and the exact defects, Section 6 the compositions of copies of the Petersen 4-pole, and Section 7 the open questions.

2 Independent certification

2.1 The graphs

We use Putman’s normalized digest: the SHA-256 of the compact JSON serialization of the sorted edge list with $u < v$. The three graphs are G_{112} (digest `dc16cc18...e8b`), H_{112} (digest `0f2d8858...35c7`), both transcribed from Putman’s CC0 archive, and G_{52} , transcribed from the appendix of [5] with vertices renumbered from 1..52 to 0..51 (digest `27db5d3b...`); the full edge lists are in the repository.

Proposition 2.1 (machine-verified). *Each of G_{112} , H_{112} , G_{52} is simple, cubic, connected, has edge connectivity 3, girth 5, and cyclic edge connectivity exactly 4: no set of at most three edges is cycle-separating (exhaustive search over all edge subsets of size at most three), and an explicit cycle-separating 4-cut exists (edges $\{2, 6, 9, 12\}$ of the sorted edge lists of both 112-vertex graphs and $\{2, 3, 8, 14\}$ of G_{52}).*

2.2 Encodings

Our Petersen encoding uses only edge-image variables $y_{e,f}$ ($e \in E(G)$, $f \in E(P)$), one image per edge, with the constraint that adjacent edges of G receive distinct adjacent edges of P . This suffices because P is triangle-free: three pairwise adjacent distinct edges of a triangle-free graph form a star. Two units break symmetry, using the edge-transitivity of P and the transitivity of an edge stabilizer on the four edges adjacent to a fixed edge. Our normal-5 encoding uses edge-color variables $x_{e,c}$, side-presence variables $p_{e,c}, q_{e,c}$ (“color c appears among the two other edges at u ”, respectively at v), and a rich indicator r_e ; richness is $\neg(p_{e,c} \wedge q_{e,c})$ for every c and poorness is $p_{e,c} \leftrightarrow q_{e,c}$ for every c . Neither encoding shares a variable scheme with Putman’s vertex-witness and missing-pair encodings.

Theorem 2.2 (machine-verified, replication). *G_{112} , H_{112} and G_{52} admit no Petersen coloring and no normal 5-edge-coloring. All six refutations, and the three Petersen refutations with symmetry breaking removed, carry DRAT proofs accepted by `drat-trim` (Table 1). The same encoders accept the Petersen graph, K_4 , the prism and the flower snarks J_5 , J_7 with checker-validated colorings. Putman’s four archived proofs are accepted by our `drat-trim` binary against his archived formulas.*

instance	variables	clauses	solve (s)	proof (bytes)	check (s)
G_{112} Petersen	2,520	73,250	65.6	98,426,474	43.7
G_{112} normal-5	2,688	11,088	1,216.8	786,818,036	900.5
H_{112} Petersen	2,520	73,250	29.5	95,632,373	24.5
H_{112} normal-5	2,688	11,088	977.2	868,156,862	1,293.5
G_{52} Petersen	1,170	34,010	1.8	4,094,201	1.2
G_{52} normal-5	1,248	5,148	178.8	226,490,830	231.8

Table 1: Certified refutations (CaDiCaL 1.7.3 under WSL, `drat-trim`); every check printed **s VERIFIED**.

3 Perfect matching covers

Theorem 3.1 (machine-verified). *Each of G_{112} , H_{112} , G_{52} admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, three perfect matchings with empty intersection, and a cover by four perfect matchings. Consequently their perfect matching index equals 4.*

Proof. The matchings are explicit (the repository file `witnesses.json` of computation EXP-002 lists them as edge index sets) and were re-verified by a checker that tests each set to be a perfect matching and then counts the coverage of every edge. The index is at least 4 because the graphs are not 3-edge-colorable (Theorem 2.2 and the trivial direction of the conjecture). $\square$

The value 4 is one below the Petersen graph’s own perfect matching index 5, which our control recovers with a verified refutation of the four-matching instance. Thus, on the first counterexamples, the Berge–Fulkerson, Berge and Fan–Raspaud conjectures all hold, and the graphs are in this sense better behaved than the Petersen graph.

4 Cycle double covers, flows, oddness and resistance

Theorem 4.1 (machine-verified). *Each of G_{112} , H_{112} , G_{52} admits a 5-cycle double cover (five even subgraphs covering every edge exactly twice) and a nowhere-zero $\mathbb{Z}_5$ -flow, and none admits a nowhere-zero $\mathbb{Z}_4$ -flow. The oddness and resistance are*

$$\text{odd}(G_{112}) = \text{odd}(H_{112}) = 4, \quad \text{odd}(G_{52}) = 2, \quad \text{res}(G_{112}) = \text{res}(H_{112}) = 3, \quad \text{res}(G_{52}) = 2.$$

Proof. Double covers and flows are explicit witnesses; the 4-flow refutations carry checked proofs. Oddness is decided from both sides: for a bound b , the formula asks for a perfect matching M and a vertex 2-coloring of the 2-factor $E \setminus M$ with at most b monochromatic edges. An odd cycle forces at least one monochromatic edge and an even cycle admits none, so the least satisfiable b is the oddness. Bounds 1, 2, 3 are refuted with checked proofs for the 112-vertex graphs and bound 4 is satisfied by a perfect matching whose 2-factor has exactly four odd cycles (counted by the checker); for G_{52} bound 1 is refuted and bound 2 satisfied. Resistance uses a deletion indicator per edge and a proper 3-edge-coloring of the remaining edges, with the same two-sided protocol. $\square$

The oddness 4 of the 112-vertex graphs is consistent with the theorem that the smallest snarks with oddness at least 4 and cyclic connectivity 4 have order 44 [8]. The 52-vertex graph, by contrast, has the least possible oddness of a snark.

5 Normal 6-edge-colorings and the Petersen defect

5.1 Normal chromatic index

Mazzuoccolo and Mkrtchyan proved that every simple cubic graph admits a normal 7-edge-coloring and asked whether 6 colors always suffice for bridgeless cubic graphs [6]; the question is restated as Conjecture 6 in [5], who verified a strong normal 6-edge-coloring (every edge rich) of one 112-vertex graph.

Theorem 5.1 (machine-verified). *Each of G_{112} , H_{112} , G_{52} admits a normal 6-edge-coloring and a strong normal 6-edge-coloring. Since none admits a normal 5-edge-coloring (Theorem 2.2), their normal chromatic index is exactly 6.*

The colorings are explicit witnesses re-verified by a checker that recomputes, for every edge, the number of colors on its two end stars. All six instances solved in well under a second.

5.2 How far from colorable: the Petersen defect

Define the *Petersen defect* of a cubic graph G as the least number of vertices at which the star condition fails, over all maps $E(G) \rightarrow E(P)$ (equivalently, the least number of vertices whose star constraints must be dropped for a coloring to exist); a Petersen coloring is a map of defect 0. Encoding the defect through a cardinality constraint over all vertices proved hard for the solver (computation EXP-004): the bound-1 instances of G_{112} and H_{112} reached a 30-minute cap without a decision, and the bound-1 instance of G_{52} was refuted in 532.4 seconds, with a verified proof. We therefore decided the defect by *designated relaxation*: for each vertex v , a formula whose star constraints are dropped at v only; and for each pair $\{u, v\}$, a formula relaxed at both.

Theorem 5.2 (machine-verified). *The Petersen defect of G_{52} is exactly 2. For every one of its 52 vertices, the map relaxed at that vertex alone does not exist (DRAT proofs checked by *drat-trim*, each in about two seconds), while every one of the 1,326 vertex pairs admits a map whose star condition fails exactly at the two relaxed vertices (explicit witnesses; the checker reports exactly those two bad vertices; the slowest instance took 0.73 seconds).*

Thus the obstruction in the smallest known counterexample is not localized: no single vertex, in particular none of the four vertices outside the six disjoint copies of F (measured in Section 6), can absorb it, whereas every pair can. The first half is not an accident of G_{52} :

Theorem 5.3 (derived). *Let G be a cubic graph and $\sigma : E(G) \rightarrow E(P)$ any map. Then the number of vertices of G at which σ fails the star condition is never exactly one. Hence the Petersen defect of a cubic graph is 0 or at least 2, and G_{52} attains the minimum possible defect of a counterexample.*

Proof. For a vertex v let $\chi_v \in \mathbb{F}_2^{E(P)}$ be the indicator of the multiset of the three labels at v (repeated labels cancel). Every edge of G contributes its label to both endpoints, so $\sum_v \chi_v = 0$. If u satisfies the star condition then χ_u is the indicator of a vertex star of P , an edge cut. Suppose v is the only bad vertex; then $\chi_v = \sum_{u \neq v} \chi_u$ is a sum of cuts, hence a cut of P , of odd weight 1 or 3. The Petersen graph is bridgeless, so the weight is 3 and the three labels at v are distinct and form a 3-edge cut $\partial(S)$. Since P is cyclically 5-edge-connected, one side of the cut, say S , induces a forest with $c \geq 1$ components, so $3 = |\partial(S)| = 3|S| - 2(|S| - c) = |S| + 2c \geq |S| + 2$ and $|S| = 1$: the cut is a star, and v satisfies the star condition after all, a contradiction. $\square$

The argument gives more: for any map, the sum of χ_v over the bad vertices lies in the cut space of P . For the 112-vertex graphs the theorem already gives defect at least 2; whether it equals 2 is decided by pair relaxations (computation EXP-006) for the pairs of vertices outside the twelve disjoint copies of F : all 120 such pairs of G_{112} and all 120 of H_{112} are critical (explicit witnesses, at most 3.1 seconds each), so both 112-vertex graphs have Petersen defect exactly 2 as well. The full sweeps show that, as for G_{52} , every one of their 6,216 vertex pairs is critical (explicit witnesses, slowest 20 seconds; the checker reports exactly the two relaxed vertices as bad in all 13,758 pair witnesses of the three graphs). We call this property, verified here for the three graphs and unexplained by the parity argument, *universal 2-criticality*.

6 Compositions of copies of the Petersen 4-pole

All known counterexamples contain the 4-pole F [5]. A packing computation (exhaustive enumeration of the 8-vertex sets with a 4-edge boundary that induce F , followed by an exact maximum disjoint packing) shows that G_{52} contains exactly six disjoint copies of F and four further vertices, and that each 112-vertex graph contains twelve disjoint copies and sixteen further vertices. This suggests the class $\mathcal{C}(k, m)$ of simple cubic bridgeless graphs made of k disjoint copies of F and m free vertices, the $4k + 3m$ semi-edges being joined by a perfect matching.

Proposition 6.1 (derived; machine-witnessed). *Every graph in $\mathcal{C}(k, 0)$ has a Petersen coloring. Hence no counterexample consists solely of copies of F .*

Proof. F admits a Petersen coloring in which its four semi-edges carry the same edge x of P (the distance-zero case of Jooker’s Lemma 2.1 [4]; an explicit such coloring is recorded in the repository). Color every copy by it, with the same x . Every join edge then carries x at both ends, and every vertex star is a star of a copy of F , hence a star of P . $\square$

The counterexample-guided search (computation EXP-005: join formula for the matching; structural check; Petersen-colorability of each candidate; sound blocking clauses learned from every coloring found, each recorded with its witness) exhausted $\mathcal{C}(5, 0)$ and $\mathcal{C}(6, 0)$ in seconds by exactly such a universal coloring, but did not converge on classes with free vertices: the 26-vertex control $\mathcal{C}(3, 2)$ stopped at a 15-minute time limit after 1,059 iterations and 29,052 learned clauses, and $\mathcal{C}(5, 2)$ (42 vertices) stopped at a two-hour time limit after 1,992 iterations and 80,832 learned clauses, both without finding a counterexample and without exhausting; $\mathcal{C}(6, 2)$ was not decided. We do not claim anything about classes the search did not exhaust; a symmetry-broken enumeration is a natural next step.

7 Scope and open questions

Nothing here bears on the general Berge–Fulkerson, cycle double cover, 5-flow or normal 6-edge-coloring conjectures: the audit concerns three graphs. The questions left open by the disproof, the smallest counterexample in [38, 52], cyclically 5-edge-connected counterexamples, and Conjecture 6 of [5], remain open. The 68-vertex graph announced on a social network is not covered.

Data availability

The edge lists, encoders, checkers, witnesses and computation manifests are in the repository https://github.com/fsantibanezleal/CAOS_RESEARCH, under `problems/combinatorics/petersen-coloring/`. An identifier EXP-*nnn* denotes the archived computation in the folder `experiments/EXP-nnn-*` of that directory, with a deterministic runner and its artifacts with SHA-256 digests. Section 2 rests on EXP-001, Section 3 on EXP-002, Section 4 on EXP-003, Section 5 on EXP-004 and EXP-006, and Section 6 on EXP-005.

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